



From words to insights: Text analysis in business research[☆]

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ABSTRACT

Business success relies on effective stakeholder communication, much of which occurs via or can be transcribed into text. Yet, business researchers often lack coherent frameworks to conceptualize business-relevant communication and its underlying logics. We thus consider business research from a message design logic lens to offer a conceptual foundation for research seeking to understand the content, style, and structure of business communication. Business researchers also underutilize modern tools for analyzing text data. Hence, our comparison of current methodologies for analyzing text (i.e., topic models, dictionaries, supervised machine learning, and large language models) points out their respective advantages, limitations, and applications. An overview of recent studies in the *Journal of Business Research* identifies how these methods are used to extract insights from business communication. We offer practical guidelines for authors and reviewers on method selection, implementation, and evaluation, and conclude by proposing future directions for business research using text data.

1. Introduction

Effective verbal communication is fundamental to business success, as it shapes how stakeholders exchange information, resolve conflicts, and influence one another. In practice, this communication occurs either through written text or transcribable audio such as emails, internal reports, social media posts, interviews, and meeting notes. These unstructured text data provide the opportunity for a wealth of research insights (Balducci and Marinova 2018); the communicator's background, intentions, and relationship to the recipient as well as the impact on receivers are all reflected in how messages are designed (Ludwig et al., 2025). Given that the availability of text data and its use in research grows, there is a need to revisit how to usefully conceptualize and analyze text-based communication to derive meaningful business insights.

Despite the recognized importance of communication for business outcomes, existing business research often struggles in two critical areas: (1) it lacks coherent conceptualizations or frameworks that help explain the underlying logics by which business communication gives insight into stakeholder characteristics, relations and behaviors, and (2) it tends

to underutilize contemporary methods for extracting insights from text data. These shortcomings limit the conceptual and methodological rigor of text analysis in business research.

To help address the first shortcoming, we draw on message and communication design logics (O'Keefe, 1988) to offer a conceptual foundation for understanding how the content, style, and structure of a verbal message reflect the communicator's background, goals, and context, and how these choices also influence audience perceptions and responses. We define communication as a purposeful design activity in which business stakeholders, whether individuals or organizations, structure, shape, and condition their communication to serve functional aims. Any communicator designs messages based on implicit logics or internal assumptions about communication (e.g., what is likely appropriate? What is likely most persuasive?), and these logics fundamentally shape the content, styles and structure of the messages they produce. This framework enables researchers to reason that the content, style, and structure of texts reliably reflect the communicator's characteristics, relationships, and intentions, and can be used to predict the potential impact of their communication.

To help address the second shortcoming, we respond to the rapid

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growth of text data by reviewing up-to-date tools for its analysis. These include the use of predefined word lists such as LIWC (<https://www.liwc.app/>) and WordStat (<https://provalisresearch.com/>), weighted dictionaries like Evaluative Lexicon 2.0 (<https://www.lexicalsuite.com/>) and VADER (<https://github.com/cjhutto/vaderSentiment>), topic models like Latent Dirichlet Allocation, supervised machine learning techniques such as Naïve Bayes and Support Vector Machines, and large language models like BERT (https://huggingface.co/docs/transformers/en/model_doc/bert), BART (https://huggingface.co/docs/transformers/en/model_doc/bart), ChatGPTs (<https://platform.openai.com/docs/models>), and LLAMAs (<https://www.llama.com/>). Each offers distinct advantages and limitations in processing and programming requirements, accuracy, and contextual sensitivity. By providing an overview of up-to-date text analysis methodologies, and discussing their key benefits, drawbacks, and measures of validity, we thus seek to aid business researchers in selecting and using appropriate tools.

Compared to existing reviews on text analysis, such as Berger et al. (2020), Feuerriegel et al. (2025), and Humphreys and Wang (2017), we contribute to business research in a number of unique ways. First, prior reviews often focus broadly on marketing or social media contexts, whereas our review is specifically tailored to the domain of business research. Second, we integrate theorizing on message design logics (O'Keefe and Lambert, 1995) and communication design logic (Aakhus, 2007) with common business research questions. This allows us to offer business researchers a conceptual rationale on how the communicator's background, situation, and intention might be reflected in a message's content, style, or structure and how these in turn influence the receiver. Third, we review and evaluate a wide range of up-to-date text analysis methodologies, including topic models, dictionaries, supervised machine learning, and large language models highlighting their respective merits, limitations, and applications. We provide examples for each method, including a summary of recent articles published in the *Journal of Business Research*. Fourth, conceptual and methodological guidelines and suggestions are provided for business researchers who want to generate insights from words, including recommendations for both authors and reviewers on how to judge the conceptual and methodological quality of text analysis research.

2. What can be explored in text data?

Text analysis in business research transforms unstructured text into numeric measures of communication-constituting concepts that reveal who is communicating, in what context, and with what intention, as well as how such communication affects the receiver (Ludwig et al., 2025). Through this lens, texts which include written and transcribed spoken communication, become valuable data sources for understanding communicators' characteristics, relational dynamics, and communicative goals. These elements are encoded in the content, style, and structure of messages, whether generated by individuals, organizations, or artificial intelligence. Because communicators differ systematically in personality, expertise, or status, and receivers interpret texts differently depending on linguistic and structural features (e.g., Herhausen et al., 2023; Ludwig et al., 2013), message design serves as both signal and strategy.

While prior work, including O'Keefe's (1988) message design logic, has examined how message composition reflects communication competence, we propose that message design logics offer broader insights into business-relevant characteristics, processes, and outcomes. Drawing from cross-disciplinary research, we advance four foundational premises that explain how communicators' background, situation, and intention manifest in their message design, and how those design choices shape recipient response. These logics provide a conceptual foundation for examining text as a lens into stakeholder identity and behavior. Rather than cataloging all possible textual elements, we present a framework that categorizes the reflective and predictive value of

verbal communication in business contexts, grounded in the idea that the way messages are constructed reflects deeper strategic, relational, and contextual meanings (see Fig. 1).

2.1. Background: Who is communicating?

The communicator, whether an individual, group, or artificial intelligence, inevitably designs the content, style, and structure of their communication through their background characteristics. Communication choices are not random but reflect the communicator's underlying beliefs, competencies, and assumptions about how communication works. According to message design logic theory, communication design reveals a communicator's level of competence and orientation toward message construction (O'Keefe, 1988). Emotion-laden language can convey sentiment and, when sufficient data is available, offer insights into speakers' characteristics (Herhausen et al., 2019). These communicative design choices thereby act as diagnostic tools, allowing researchers to infer aspects of the communicator's personality, perceptions, and attitudes by analyzing text content, style, and structure.

In business research, text analysis offers a non-intrusive method for accessing stakeholder backgrounds and perspectives. For example, Colladon et al. (2023) demonstrate that innovators tend to write more extensively and use complex language to frame novel ideas. Garner et al. (2022) show how machine learning applied to textual data can reveal consumer sentiment and dimensions of well-being. Kumar and Srivastava (2022) examine how corporate social responsibility discourses evolve over time by analyzing shifts in emphasis on their constituent dimensions. Mangiò et al. (2023) uncover how journalists co-construct brand meanings for family businesses, whether intentionally or not.

Together, these findings underscore that communication design encodes the communicator's characteristics, mental models, and priorities, making textual data a valuable resource for examining the background of individuals and organizations.

2.2. Situation: In which context does the communication take place?

Communication always occurs within a context, and the communicator's interpretation of that context informs how messages are designed, beliefs conveyed, and goals pursued. Foundational perspectives in semiotics (Kress et al., 1997), rhetoric (Gill and Whedbee, 1997), and ethnography (Gumperz and Levinson, 1991) all stress the inseparability of context and communication. Communication logics, particularly in conversational settings, reveal social and situational realities. Contextual factors can both reflect and influence communication practices, offering insights into how participants perceive relationships, adhere to context-specific norms, and navigate situational constraints (Bateson, 1972).

Empirical findings illustrate these dynamics. Cleveland, Laroche, and Papadopoulos (2015) show that language learning grants access to cultural values and interaction norms. Kacewitz et al. (2014) find that individuals in subordinate positions tend to use more first-person singular and fewer first-person plural pronouns. Kronrod and Danziger (2013) demonstrate that figurative language use differs depending on whether consumers describe hedonic or utilitarian experiences. Snefjella and Kuperman (2015) reveal that people adjust their use of abstract or concrete language based on perceived geographic, temporal, and social distance.

Therefore, communication also encodes a contextual logic that emerges from, and contributes to, the social environment or context in which it unfolds. Recognizing this contextual embeddedness enables a deeper understanding of communicative meaning and provides a theoretical anchor for researchers studying how stakeholders view and relate to their context.

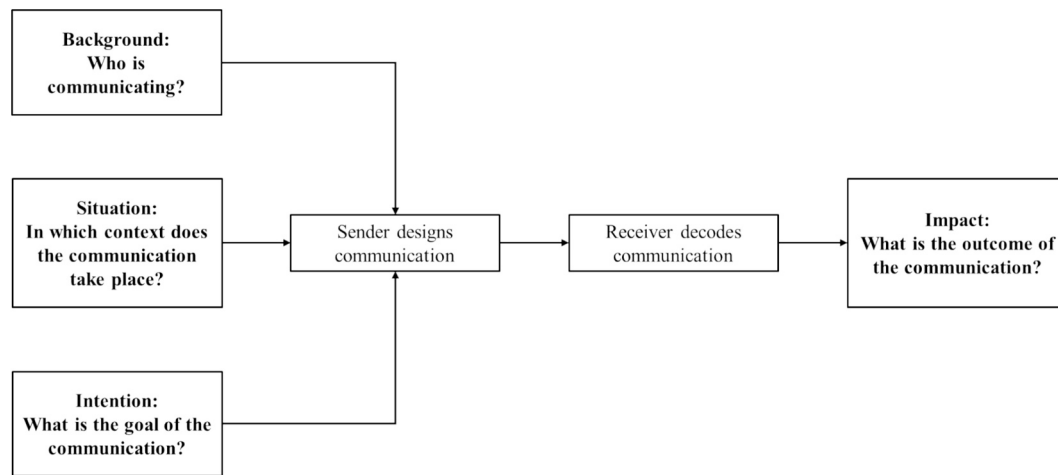


Fig. 1. Communication design logics.

2.3. Intention: What is the goal of the communication?

Communication is inherently goal-driven (Caughlin et al., 2008), and communicators' intentions are embedded in their message design choices, including content, style, and structure (O'Keefe and Lambert, 1995). Theories of speech acts, for example, conceptualize elements of language as illocutionary acts that reflect and transmit the speaker's intent (Searle, 1995). Word choices, semantic constructions (e.g., "happy," "not happy"), coherence, and adaptive message design choices (e.g., accommodating in the style of communicating) all signal communicative goals. Intentions also influence the sincerity of assertions; irony, sarcasm, and deception emerge when truthfulness is deliberately suspended, often to entertain or mislead (Meibauer, 2005).

O'Keefe (1988, p. 82) notes that communicators derive means to serve communicative goals, even if these goals are not explicitly defined or consciously recognized. This implies that variability in messages can reveal underlying differences in intentions. Empirical studies support this view. Ludwig et al. (2016) identify deceptive intentions in B2B email communication through linguistic markers. Netzer, Lemaire, and Herzenstein (2019) find that borrower narratives can signal repayment intentions. Ray, Bala, and Rana (2021) demonstrate that emotional tone, especially high-arousal negative or low-arousal expressions, affects customer brand attitudes. Villaroel Ordenes et al. (2019) use speech act theory to classify firms' communication according to underlying intentions.

These findings reinforce the view that communication is goal-oriented and that intentions are encoded in language use at multiple levels. Recognizing the intentional dimension of communication offers a conceptual basis for inferring communication goals from text data.

3. Impact: What is the outcome of the communication?

A message design also directs attention to specific interpretations while omitting others. From a constructionist view, reality is socially constructed through symbolic exchange, with communicators simultaneously producing and interpreting messages (Morgan and Smircich, 1980). Content, stylistic, and structural choices evoke associations through connotation, prompting recipients to perceive the world through particular frames. Message design logic can therefore influence how audiences perceive and respond messages.

Communicated messages act as positioned representations that foster distinct conceptualizations by the receiver of relations, intentions, and reactions (Fairclough, 1992). Put differently, message design choices create metacommunicative frames that guide receivers' situational understanding (Bateson, 1972) and influence collective interpretations and public discourse (Charland, 1987). Similarly, Searle (1976) identifies

perlocutionary acts as communicative moves that provoke behavioral change. For example, macro-level speech acts affect how people cognitively process messages, influencing attention, comprehension, and memory (van Dijk, 1997; Naveh-Benjamin and Ayres, 1986). The ability to tailor message design to the audience and purpose strengthens the potential for achieving intended outcomes, which highlights the interdependence between message designs and their effects.

Empirical studies support this view. Herhausen et al. (2023) show that empathetic complaint responses increase customer gratitude. Holmqvist, Van Vaerenbergh, and Grönroos (2017) emphasize the importance of managing language throughout service interactions. Lee and Theokary (2021) find that superstar influencers enhance engagement by leveraging emotional contagion and stylistic cues. Ludwig et al. (2022) demonstrate that freelancers respond more positively to buyers who provide moderately detailed and concrete task descriptions.

Therefore, communication, and the design of messages, is performative and outcome-driven. By studying message constructions, researchers can assess and predict their likely impact on receivers' perceptions, emotions, and behaviors.

4. How can text data be explored?

Most text analysis can be categorized into four primary approaches: topic models, dictionaries, supervised machine learning, and large language models. Table 1 summarizes the strengths, limitations, and applications of these different text analysis methods.

4.1. Topic models

Topic models aim to uncover latent themes in collections of documents, offering insight into textual patterns beyond manual inspection. Latent Dirichlet Allocation (LDA) remains a foundational probabilistic model in this space (Blei et al., 2003). LDA assumes that documents are mixtures of topics, and topics are distributed over words. Through a generative process, LDA infers the composition of topics that best characterize a document collection. For instance, Culasso et al. (2023) apply LDA to identify thematic responsibilities in the role of chief digital officers. LDA can also be usefully applied to account for (dis)similarities between different texts in a regression analysis, for example (Ludwig et al., 2022).

Despite its utility, LDA presents several limitations. Its output depends heavily on hyperparameters such as the number of topics and Dirichlet priors, and its results can be unstable due to the order effect in document processing (Abdelrazek et al., 2023; Wu et al., 2024a). LDA results are also non-theoretical and it is difficult to assign a broader conceptual category to an LDA generated topic. To address these issues,

Table 1
Overview of text analysis methods.

Method	Approach	Description	Application	Examples	Strengths	Limitations
<i>Topic Models</i>	Latent Dirichlet Allocation (LDA)	Identifies topics in a text by clustering words that frequently occur together	Topic discovery, document classification	Culasso et al. (2023); Huang et al. (2018)	Simple to implement, interpretable results	May struggle with complex datasets
	Extended LDA	Variants of LDA that incorporate additional information	Enhanced topic discovery, better document categorization	Liu and Toubia (2018); Zhong and Schweidel (2020)	More accurate topic modeling, can incorporate domain knowledge	More complex to implement and interpret
	Neural Topic Models	Uses neural networks to identify topics in text	Deep topic analysis, advanced document clustering	Büschken and Allenby (2020); Hou et al. (2023)	Handles large datasets, captures complex patterns	Requires larger amounts of data, computationally intensive
<i>Dictionaries</i>	Basic Dictionaries	Uses pre-existing dictionaries to identify and classify text (elements)	Basic sentiment analysis, keyword extraction	Ludwig et al. (2013); Rocklage and Fazio (2020)	Easy to use, interpretable results, simple complexity, low data requirement	Limited to the scope of the dictionary, may miss nuanced meanings, limited scalability and adaptability
	Refining Dictionaries	Enhances basic dictionaries with additional terms and context	Improved sentiment analysis, basic context-aware text classification	Cho et al. (2024); Herhausen et al. (2019)	More accurate classifications than basic dictionaries, moderate performance	Still limited by predefined terms, may require frequent updates, limited scalability and adaptability
	New Dictionary Development	Creates custom dictionaries tailored to specific applications	Domain-specific text analysis, specialized sentiment analysis	Homburg et al. (2020); Singh et al. (2020)	Highly accurate for specific domains, highly interpretable	Time-consuming to develop, requires domain expertise, limited scalability and adaptability
<i>Supervised Machine Learning</i>	Classifiers	Models trained on labeled data to classify or predict text	Text classification, spam detection, sentiment analysis	Hartmann et al. (2019); Zhang et al. (2023)	High accuracy with sufficient labeled data, good scalability and adaptability	Requires large amounts of labeled data, may not generalize well, less state-of-art on very nuanced tasks
<i>Large Language Models</i>	Encoder-only models	Analyze text by creating contextualized word representations. Process input bidirectionally for better understanding.	Text classification, named entity recognition	Devlin et al. (2019); Shankar and Parsana (2022)	Strong text comprehension, effective at contextual word representation	Require task-specific fine-tuning for different applications, not inherently interpretable
	Decoder-only models	Generate text by predicting the next token based on prior context	Text generation, dialogue systems, content completion	Brown et al. (2020); Reisenbichler et al. (2022)	Fluent and context-aware, adaptable to various tasks	Sensitive to prompt phrasing, can be costly; lacks transparency
	Encoder-decoder models	Process input text into a contextualized representation with an encoder and generate output text with a decoder.	Machine translation, text summarization	Arora, Chakraborty and Nishimura (2025); Lewis et al. (2019)	Balance of comprehension and generation	Computationally expensive

several LDA extensions have been introduced.

Guided LDA incorporates external knowledge to prioritize relevant topics (Toubia et al., 2019), allowing researchers or firms to direct topic discovery toward strategic areas of interest. Seeded LDA uses predefined seed words to guide the model, offering more coherent and focused results, particularly useful in domains where expected topics are known (Watanabe et al., 2024). Hierarchical LDA builds layered topic structures, enabling analysis at multiple levels of granularity (Griffiths et al., 2003). Zhong and Schweidel (2020) use these extensions to monitor shifts in social media discourse. While these variants improve flexibility and focus, they also introduce trade-offs. Guided LDA depends on the quality and relevance of external inputs, seeded LDA on seed word selection, and hierarchical LDA on computational complexity and structural assumptions.

Neural Topic Models (NTMs) extend topic modeling using neural network architectures. These models generate latent document representations through deep learning, offering more flexible and context-aware topic extraction (Wu et al., 2024b). BERT and other transformer-based models provide deep contextual embeddings that can be adapted for topic modeling. For example, BERTopic combines BERT with clustering and dimensionality reduction to identify coherent topics. Büschken and Allenby (2020) demonstrate that NTMs improve analysis by capturing relationships between sentence-level structures.

Key advantages of NTMs include the ability to jointly train with other models, optimize topic coherence, and scale effectively for large datasets. These features make NTMs particularly useful in digital marketing, social media monitoring, and content automation. However, NTMs are less interpretable than LDA. Their reliance on dense latent representations makes it difficult to trace how input data leads to specific topic

outputs, and they often struggle with sparse data. This can lead to information loss, especially in domains with rare or domain-specific terminology (Abdelrazek et al., 2023; Wu et al., 2024b).

4.2. Dictionaries

Dictionaries classify text using predefined word lists linked to specific categories or sentiments (Taboada et al., 2011). A common approach involves applying established dictionaries such as Linguistic Inquiry and Word Count (LIWC), valued for its comprehensiveness and ease of use (Tausczik and Pennebaker, 2010). For instance, Ludwig et al. (2013) use LIWC to quantify emotionality in customer reviews, enabling researchers to assess psychological and social patterns in text. While especially effective across communication contexts, dictionaries often miss domain-specific nuance, limiting their precision in specialized applications.

To address this, researchers tend to refine or expand existing dictionaries by updating word lists to better reflect specific industries or linguistic trends. This may involve adding relevant terms, removing outdated ones, or adjusting word weights. Herhausen et al. (2019), for example, refine LIWC to capture consumer emotions in social media firestorms more accurately.

When no suitable dictionary exists, a new one can be developed to align with a specific construct. This approach, while almost as resource-intensive as a scale development paper, enables precise measurement of novel or context-specific phenomena. Singh et al. (2020), for example, create dictionaries to capture influence tactics in B2B negotiations.

Dictionaries offer several practical advantages. They are easy to implement, interpretable, computationally efficient, and do not require

labeled training data or programming skills. These features make them suitable for large-scale applications and environments where transparency is essential. However, their limitations in terms of accuracy are significant. Dictionaries rely on static word lists, often failing to account for linguistic ambiguity, synonymy, or shifting vocabulary. Words like “mad” can vary in meaning by context, and dictionaries cannot adapt without manual updates. Their effectiveness depends heavily on coverage and quality, and biased or incomplete dictionaries may lead to misleading conclusions.

Despite their simplicity, dictionaries remain valuable tools when carefully selected, refined, or constructed for the task at hand. They provide interpretable insights into text data and serve as a foundational method in both exploratory and applied business research.

4.3. Supervised machine learning

Supervised machine learning (SML) addresses several limitations of dictionary-based methods by learning from labeled datasets. In SML, each data point consists of an input text and an output label, and the objective is to train a classifier that predicts labels for new, unseen data. To enable this, textual data must be transformed into numerical features, typically through techniques such as Bag-of-Words, which represents documents as vectors of word frequencies, or N-grams, which capture local word sequences to preserve limited contextual information.

SML models are widely used for text classification tasks, including sentiment analysis, spam detection, and topic categorization. Common algorithms include logistic regression, which estimates class probabilities using a logistic function; Naive Bayes, which applies probabilistic reasoning under independence assumptions; and support vector machines (SVMs), which identify optimal boundaries between classes in high-dimensional feature spaces (Hartmann et al. 2019). Beyond classification, SML supports sequence labeling tasks such as named entity recognition (NER) and part-of-speech (POS) tagging. In NER, entities like organizations, locations, or monetary values are identified and categorized within the text. For example, in the sentence “Google is looking at buying a startup for \$100 million in the U.K.,” NER labels “Google” as an organization, “U.K.” as a location, and “\$100 million” as a monetary value. POS tagging assigns grammatical roles such as nouns, verbs, or adjectives to each word, enabling syntactic analysis and interpretation. Both tasks require annotated data for training, where models learn to predict structured outputs from labeled sequences.

Despite their flexibility, traditional SML approaches have constraints. They rely heavily on annotated training data, making them unsuitable for unlabeled datasets (Alslaity and Orji, 2024). These models are often task-specific and cannot be easily adapted across different domains or objectives (Sarkar and Etemad, 2020). Moreover, their performance depends on the availability of large, high-quality labeled datasets, which may not always be accessible.

Nevertheless, SML remains foundational in text analysis due to its interpretability, reproducibility, and potential for more accurate classifications of constructs in texts.

4.4. Large language models

Large language models (LLMs) advance text analysis by addressing key limitations of supervised machine learning. Models such as GPT,

LLAMA, DeepSeek, BERT, and RoBERTa, built on transformer architectures, leverage deep learning and large-scale pretraining to process language with high fluency and contextual sensitivity. Pretrained on massive corpora from diverse sources, these models acquire broad linguistic competence and can be fine-tuned for specific tasks such as sentiment analysis or classification (Jiang et al., 2023; Liao et al., 2023).¹

Transformers rely on self-attention mechanisms to model long-range dependencies within text (Vaswani et al., 2017), enabling LLMs to handle complex language tasks, including aspect-based sentiment analysis and syntactic ambiguity, with high accuracy. A key advancement of LLMs is their ability to perform zero-shot and few-shot learning. Without task-specific training, models can fairly accurately infer correct classifications by drawing on their pretraining. For example, given the prompt “Classify the sentiment of this review: ‘I love this brand! Their products never disappoint,’” an LLM will mostly return a “Positive” classification without requiring labeled examples for training. Few-shot learning builds on this by providing minimal examples to improve task performance through in-context learning. Of course, training LLMs on labeled example data will help to further improve their accuracy.

LLMs follow three main architectural formats. Encoder-only models like BERT generate contextualized representations for tokens, useful in classification and tagging tasks. Sousa et al. (2019) apply BERT to extract sentiment from financial news and predict Dow Jones movements. Decoder-only models like GPT specialize in text generation and continuation by predicting subsequent words. Goli and Singh (2024) use GPT-3.5 and GPT-4 to simulate human survey responses for preference data. Encoder-decoder models combine the strengths of both structures and are suited for tasks such as translation and summarization. For example, BART integrates bidirectional encoding with autoregressive decoding, enabling coherent output grounded in context. Kanungo et al. (2022) employ BART to generate optimized ad headlines.

While LLMs offer scalability, contextual precision, and generalization capabilities unmatched by earlier models, researchers are advised to use them with care and ensure their classification is valid by comparing outcomes with human coding (Li et al., 2024; Ye, Yoganarasimhan and Zheng, 2025). Moreover, researchers will need to think about ensuring the reproducibility of their LLM use in future research by detailing prompts and versions of the LLM used. LLMs are mostly not private and therefore sensitive text analysis cannot be performed through publicly available LLMs. The costs of running them may also sometimes preclude researchers from using them.

5. Text analysis in the Journal of Business Research

To explore the state of text analysis in the *Journal of Business Research* regarding design logics and methods used, we use the search query “text analysis” OR “text mining” for all articles published from Volume 162 (July 2023) until Volume 179 (June 2024). This query resulted in 61 articles, nine percent of all articles published during this time. With manual inspection, we excluded editorials, literature review articles, and qualitative research, and all articles without a communication focus (e.g., CEO backgrounds’ impact on firms’ green initiatives, and the effect of digitalization on gender equality). We kept the 20 articles where text data has been analyzed with quantitative methods to study business communication. These articles are summarized in Table 2. One article

¹ Embeddings and LLMs share some common features, e.g., the underlying transformer architectures and extensive pretraining on large text corpora. However, they are optimized for different tasks. LLMs focus on “understanding” and generating coherent and contextually-relevant text based on input prompts while embeddings focus on producing high-quality embeddings for downstream tasks, fundamental to how LLMs process language. Compared to bag-of-words approaches such as dictionaries, embeddings more precisely represent the semantic meaning of words and their interrelations with other words.

Table 2

Diversity of text analysis in the Journal of Business Research.

Article	Context	Sender	Receiver	Main Construct	Design Logic	Method	Main Insight
Dhakate & Joshi (2023)	E-healthcare services	Patients (36,468 reviews)	343 Doctors	Sentiment	Situation	LLMs (Azure Machine Learning), Supervised Machine Learning (Part of Speech), Dictionaries (NRC sentiment, JMP Pro)	Patients' sentiment reveals their expectations from online healthcare providers.
Filieri et al. (2023)	Peer-to-peer marketplaces	Customers (14,680 reviews)	468 Airbnb listings	Experiences	Situation	Supervised Machine Learning (Leximancer)	Consumers have differentiated cognitive, symbolic, emotional, and aesthetic experiences when evaluating standard versus premium peer-to-peer offerings.
Guo et al. (2023)	Digital transformation	Firms (13,571 firm-year observations)	Investors	Digital transformation	Impact	Dictionaries (new dictionary, pre-existing dictionaries)	Digital transformation can significantly increase total factor productivity but may decrease firm performance.
Hajek et al. (2023)	Online reviews	Customers (21,000 reviews)	Customers	Sentiment	Intention	Topic Models (Word2Vec LDA), Supervised Machine Learning (Part of Speech, Flesch reading ease, Gunning FOG formula, counts of characters, words, and sentences, and word-vectors)	Product category and the verified purchase attribute are fundamental for detecting fake reviews.
Hamby et al. (2024)	Media content	Media (2,892 news articles)	Readers	Narrativity	Intention	Dictionaries (LIWC)	Narrative content and discourse features differ between real and fake news and affect content-sharing behavior.
Jiang et al. (2023)	SME websites	SMEs (1,400 loan listings)	Creditors	Information	Impact	LLM s(BERT embeddings), Topic Models (K-means++ clustering)	Content-based information (depth and dynamic metrics) in borrowers' official websites conveys SME performance.
Lee et al. (2023)	Book sales	Booksellers (276 books)	Readers	Consumer needs	Impact	Topic Models (LDA)	Growth, relational, and cognitive needs influence the longevity of bestsellers.
Liao et al. (2023)	B2B livestreaming	Live streamers (915 brand firms)	Firms	Competences	Impact	LLMs (Baidu Cloud Chinese Semantic Sentiment Analysis System), Dictionaries (new dictionaries)	Cognitive and social competencies of live streamers have a positive effect on sales performance, while emotional competence has a negative effect.
Lu et al. (2023)	Online platforms	Customer reviews (14,652 reviewers)	Firms	Sentiment	Impact	Topic Models (non-negative matrix factorization), Dictionaries (TextBlob)	Weighted service quality approach outperforms average sentiment methods, with social media providing superior value in capturing customer perceptions.
Pan et al. (2023)	Environmental performance and firm response	190 SMEs (901 observations)	Investors	Readability	Situation	Supervised Machine Learning (Readability based on Word2Vec)	Environmental performance feedback has an inverted U-shaped relationship with the readability of CSR reports.
Pan, Wang, et al. (2023)	Platform economy	Firms (2,838 listed companies)	Investors	Technological innovation	Impact	Dictionaries (new dictionary)	Platform economy improves firms' enterprise value, which is mediated by the technological innovation ability.
Patel et al. (2023)	Government intervention	Firms (91 firms)	Investors	Complexity and intangibility	Impact	Dictionaries (pre-existing dictionary)	Firms with higher complexity and intangibility realize positive returns after the introduction or passing of privacy laws.
Wang et al. (2024b)	Firm value	310 Executives (759,465 tweets)	Investors	Emotions	Impact	LLMs (BERT)	Emotions of executives' social media posts are associated with firm value.
Wang et al. (2024a)	Peer-to-peer lending	Borrowers (127,044 loan requests)	Lenders	Politeness	Impact	Dictionaries (LIWC, new dictionary)	Usage of polite language is positively related to likelihood of funding success.
Setten and Chen (2024)	Children's media	Franchises (78 toy and media franchises)	Young consumers	Emotions and drives	Situation	Dictionaries (LIWC)	Based on gender, emotional tones and drives are different in children's media.
Shan et al. (2024)	Online health Communities	50,636 Reviewers (93,774 reviews)	Patients of 8,472 physicians	Electronic Word of Mouth	Impact	Supervised Machine Learning (readability, comprehensiveness)	Quality and quantity of electronic Word-of-Mouth in online health communities affect patients' choices.

(continued on next page)

Table 2 (continued)

Article	Context	Sender	Receiver	Main Construct	Design Logic	Method	Main Insight
Sun et al. (2024)	Digital transformation	1,195 Firms (18,431 observations)	Investors	Digital transformation speed	Impact	Dictionaries (new dictionary)	Digital transformation speed has a U-shaped relationship with firm financial distress.
Syrdal et al. (2023)	Influencer affiliate Marketing	Affiliates (2,040 affiliates posts)	Consumers	Linguistic styles	Impact	Dictionaries (LIWC)	Linguistic features in affiliate posts drive central or peripheral processing, which in turn influence consumer engagement positively or negatively depending on style.
Wu et al. (2024a)	Online reviews	Consumers (11,580 reviews)	Hotels	Service attributes	Impact	Supervised Machine Learning (TF-IDF algorithm), LLM (PaddlePaddle)	Online reviews can be mined to extract competitive factors dynamically; the hybrid method outperforms traditional approaches.
Yoganathan & Osburg (2024)	Conversational agents	Consumers (4,995 reviews)	Consumers	Agency and experience	Impact	Dictionaries (LIWC, pre-existing dictionary), Topic Models (LDA)	Perceiving voice-based conversational agents as having emotional experience – whether as moral agents or patients – increases user satisfaction, especially among highly innovative users.

uses topic models, eight articles use dictionaries, three articles use supervised machine learning, one article uses LLMs, and seven articles use multiple methods to extract insights from text data. This review underscores the diversity of design logics and methods used by authors in the *Journal of Business Research*.

6. How can the different text analysis methods be implemented?

In the following, we provide recommendations on how to implement the different methods, as summarized in Fig. 2, and how to evaluate the methodological rigor of different methods.

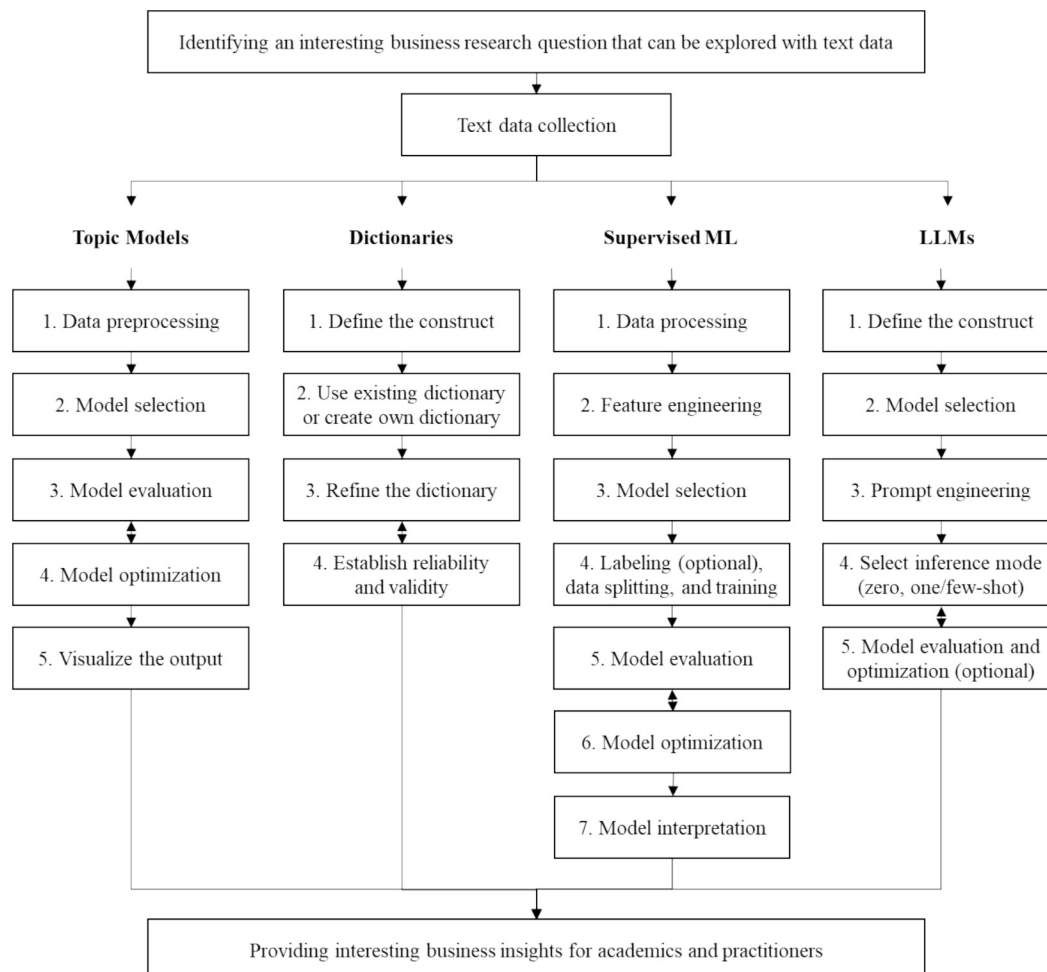


Fig. 2. Implementing the text analysis methods.

6.1. Implementing topic models

Step 1: Data preprocessing. Once a relevant dataset is identified, preprocessing ensures the text is clean and consistent (Büschken and Allenby, 2020). This typically involves removing HTML tags (e.g., <div>, <a href>), special characters (e.g., @, #), rare words (<1% of the text), and irrelevant numbers. Text is converted to lowercase, punctuation and stop words (e.g., “and”, “the”) are removed, and lemmatization reduces words to their base forms (e.g., “running”, “ran”, “runs” become “run”). Missing values are either dropped or imputed.

Step 2: Model selection. Topic modeling methods range from traditional models like LDA to advanced approaches such as BERTopic.² LDA requires selecting parameters like the number of topics (K) and Dirichlet priors, often through iterative experimentation and back-and-forth refinement of the model (see steps 3 and 4). BERTopic, in contrast, uses pre-trained embeddings (e.g., Word2Vec,³ GloVe,⁴ BERT) and clustering algorithms such as HDBSCAN⁵ or K-Means⁶ (McInnes et al., 2017). Embeddings convert text into high-dimensional semantic vectors, which are clustered to form topics. This approach leverages the more advanced transformer-based language models over traditional Bag-of-Words approaches, to identify closely related clusters. In both approaches, key terms are extracted from each cluster to define interpretable topics, providing a concise linguistic label for each identified higher-order topic.

Step 3: Model evaluation. Evaluation ensures the quality and relevance of discovered topics. *Coherence* metrics like CV (Röder et al., 2015), UMass (Mimno et al., 2011), and NPMI (Bouma et al., 2009) measure the semantic consistency of top topic words. *Stability* assesses how consistently the model produces similar topics across runs with different initializations (Abdelrazek et al., 2023), helping detect randomness in output. *Perplexity* evaluates how well traditional models like LDA predict unseen data (Wu et al., 2024a), although it may not correlate with human interpretability. *Human evaluation* involves expert review of topics for contextual sense and practical relevance, often revealing subtleties missed by metrics.

Step 4: Model optimization. Model refinement involves adjusting the number of topics, LDA hyperparameters, or iterations to balance specificity and generality. For more advanced models like BERTopic, this includes modifying embedding models and clustering settings. Visualization tools assist in identifying incoherent or overlapping topics. Expert feedback during this phase ensures interpretability and contextual alignment. The evaluation and optimization process is iterative, adjusting parameters and assessing model fit to ensure meaningful topic structures and prevent the overfitting of models (e.g., too many topics, sparse hyperparameters, or limited corpus sets).

Step 5: Visualize the output. While topics are usually difficult to define conceptually, some visualizations may help with the topic interpretation if that is necessary for the research objectives. *Word clouds* display the most frequent words per topic, with size indicating prominence. *Inter-topic distance maps* (e.g., pyLDAvis, BERTopic) visualize relationships between topics as spatial distances, with circle size indicating prevalence. *Topical networks* depict connections between topics as nodes and edges, revealing structural relationships. *Topic distribution heatmaps* show topic prevalence across documents, highlighting variability or concentration. *Hierarchical visualizations* uncover subtopics within broader themes, offering detailed insights into topic structure. Carefully visualizing reduced dimensions allows for interpretable and representative topic labels.

6.2. Implementing dictionaries

Step 1: Define the construct. The process begins with a clear definition of the construct(s) to be measured. For instance, in developing a dictionary for sentiment, it is essential to specify what qualifies as positive, negative, or neutral sentiment within the specific communication context from where the texts should be mined. Similarly, when studying emotions, the focus is ideally narrowed to relevant emotions in a specific context. A well-defined construct ensures conceptual clarity of what the dictionary will be used for and guides all subsequent steps.

Step 2: Use existing dictionary or create own dictionary. With a defined construct, and if no existing dictionary is available, an initial list of conceptually relevant terms, words, and expressions needs to be assembled. This involves consulting existing dictionaries, thesauri, domain-specific lexicons, and prior research to gather synonyms, related expressions, and context-specific terms. Terms should also include specific expressions for a context to reflect the domain accurately and enhance coverage. For example, ‘sick’ in a health context means something else than ‘sick’ in a gaming context. Yet, to capture positive sentiment in a gaming context, the word ‘sick’ is a great identifying term.

Step 3: Refine the dictionary. The preliminary list (or existing dictionary) is then refined through expert evaluation. Subject matter experts independently assess each term for relevance. Inter-rater reliability metrics such as Cohen’s Kappa or Krippendorff’s Alpha are used to quantify agreement. A high reliability score confirms consensus, while lower scores should prompt re-evaluation or post-discussions amongst the coders. Words consistently judged as relevant are retained, while irrelevant or ambivalent terms are removed, increasing the dictionary’s precision in detecting the construct (while potentially also increasing the chances of missing relevant mentions because the terms are not always and only sometimes relevant).

Step 4: Establish reliability and validity. The refined dictionary is applied to a representative text sample to assess its measurement accuracy. Convergent validity is tested by examining whether different indicators of the construct align, while discriminant validity ensures the dictionary does not overlap with unrelated concepts. Precision, recall and F1-scores should be reported to confirm that the dictionary captures the intended construct accurately.

Based on validation results, the dictionary is further refined to reflect new insights or changes in language use. Terms may be added, removed, or adjusted to maintain relevance. This feedback loop ensures ongoing refinement and sustained applicability in dynamic linguistic and contextual environments. More detailed recommendations and best practices on implementing (new) dictionaries are provided by Marinova et al. (2018).

6.3. Implementing supervised machine learning

Step 1: Data processing. Preprocessing begins with cleaning the text by removing noise, tokenization, handling missing values, normalizing case, stripping punctuation and stop words, and removing non-textual attributes such as emojis, and applying stemming or lemmatization. The quality of preprocessing the data directly impacts the model’s performance and allows for more accurate training and labeling.

Step 2: Feature engineering. Raw text must be converted into numerical representations suitable for machine learning algorithms. Common techniques include *TF-IDF*, which weights word frequency against its rarity across documents, highlighting important terms. For instance, “excellent” might receive a high TF-IDF score if it is frequent in a specific review but rare elsewhere; *Word embeddings* like Word2Vec or GloVe map words into continuous vector spaces, preserving semantic relationships. For example, “king” and “queen” appear close together in vector space. These embeddings capture contextual meaning and improve model accuracy.

Step 3: Model selection. Appropriate models are selected based on the

² <https://maartengr.github.io/BERTopic/index.html>.

³ <https://www.tensorflow.org/text/tutorials/word2vec>.

⁴ <https://nlp.stanford.edu/projects/glove/>.

⁵ https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html.

⁶ <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>.

problem and dataset characteristics (e.g., Hartmann et al. 2019, Hartmann et al., 2023). Examples include *Naïve Bayes*, a probabilistic classifier, which assumes feature independence and is efficient for text classification (Webb, Keogh and Miikkulainen, 2010), *Support Vector Machines* (SVMs) which performs well in high-dimensional settings by finding the optimal separating hyperplane (Ghaddar and Naoum-Sawaya, 2018), and *Logistic regression* which efficiently estimates probabilities for binary classification problems (Allison, 2012).

Step 4: Labeling (optional), data splitting, and training. The chosen model is trained on labeled data, where input features are paired with target values. Labels may come from existing sources, such as Yelp’s filtering algorithms or helpfulness votes on reviews, which indicate perceived value. When no labels exist, manual annotation is required. Manual labeling follows five steps: a. Define clear labeling criteria (e.g., certainty levels in reviews); b. Train coders to apply these criteria; c. Coders label data accordingly; d. Measure inter-coder reliability to ensure consistency; e. Refine labeling and retrain the model in iterative cycles. For example, to classify sentiment as “positive,” “negative,” or “neutral,” human coders label a subset, which is then used to train and improve the model through repeated refinement. The labeled data is then split into training, validation and test samples. Traditionally, 80 % of the data is used for training (sometimes a separate validation set is created within the training sample for model refinement), while the remaining 20 % is reserved for testing (i.e., a holdout data that is not used for training and validation and exclusively used for the final assessment of the model’s ability to generalize).

At this point the selected model is trained on the training sample and initial hyperparameters are set (i.e. learning rate, number of epochs, batch size etc.). The choice of hyperparameters impacts how well the model learns, prevents overfitted models and how long the training takes. These hyperparameters serve as a baseline to evaluate the model and are adjusted and fine-tuned after initial evaluation (see step 6).

Step 5: Model evaluation. The trained model based on the training data is evaluated using the validation data on standard metrics for classification performance. These metrics, summarized in Table 3, assess accuracy, precision, recall, and F1-score, offering insights into how well the model performs. In this step also different models can be tested against each other (e.g., by comparing the performance of Naive Bayes, SVMs, and Logistic regression).

Step 6: Model optimization. Based on evaluation results, the model is fine-tuned by adjusting (model-specific) hyperparameters, enhancing feature engineering, or retraining on updated data. Key hyperparameters include the number of epochs (how many full passes through the data), the batch size (how many samples per weight update), the learning rate (how aggressively weights change), and the strength of L1 or L2 regularization (how firmly model complexity is penalized). Each setting involves a trade-off: a very high learning rate may overshoot optimal weights, whereas a very low rate can slow down training tremendously; too many epochs risk overfitting by memorizing noise, while too few leave the model under-trained.

Monitoring the loss curve (plotting the mean squared error and epochs) as a visual aid, allows one to inspect the tradeoff between optimization and overfitting. Alternative settings such as the early-stopping routines help by halting training the moment extra epochs no longer yield sufficient increases in performance. An iterative optimization process with model evaluation should improve robustness, interpretability, and predictive performance. Once the model is validated, the final model should be applied to the test sample to provide an unbiased evaluation of its performance. If the model also fits the test data well, minimal overfitting has taken place. A much better fitting of the training data as opposed to the test data points to overfitting.

Step 7: Model interpretation. After optimization and application of the model to the test sample, the focus shifts from how well the model performs to why it predicts this way. Post-hoc explainable AI algorithms such as LIME or SHAP (Gramegna and Giudici, 2021) can identify the most predictive words for the classifications and how the decisions in the

Table 3
Model evaluation in machine learning-based methods.

Metric	Formula	Definition	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	The ratio of correctly predicted instances to the total instances.	Useful when the classes are balanced. It can be misleading with imbalanced datasets. A model with 80 % accuracy correctly predicts 80 out of 100 instances. Depending on the field, accuracy between 70 %-90 % are acceptable.
Precision	$\frac{TP}{TP + FP}$	The ratio of correctly predicted positive observations to the total predicted positives.	Important when the cost of false negatives is high. A precision of 0.75 indicates that 75 % of the positive predictions made by the model were correct.
Recall	$\frac{TP}{TP + FN}$	The ratio of correctly predicted positive observations to all the actual positives.	Important when the cost of false positives is high. A recall score of 0.85 means the model correctly identifies 85 % of all positive instances.
F1-score	$\frac{2 * (Precision * Recall)}{Precision + Recall}$	The harmonic mean of Precision and Recall.	Balances precision and recall. Useful when there is a need to balance false positives and false negatives. An F1-score of 0.80 indicates a balanced performance between precision and recall, with higher scores indicating better overall performance.
AUC-ROC	<i>AUC measures the entire two-dimensional area under the ROC curve.</i>	Measures the model’s ability to distinguish between classes.	Useful for a broad evaluation across different classification thresholds. AUC = 1 represents a perfect model (all predictions correct), AUC = 0.5 corresponds to random guessing, and AUC < 0.5 indicates worse performance than random guessing.

Note: TP = true positive, TN = true negative, FP = false positive; and FN = false negative.

model are made, to make the obtained results interpretable for the researcher.

6.4. Implementing large language models

Step 1: Define the construct. Begin by clearly defining the construct to be measured and its contextual nuances. LLMs, unlike traditional models, work well with raw or minimally processed text, though it remains useful to remove irrelevant content, duplicates, and formatting inconsistencies. For domain-specific tasks, curating relevant data improves performance. Notably, a portion of the data should be reserved for benchmarking when evaluating model output against reference

labels in step 5.

Step 2: Model selection. Model choice depends on the task. Encoder-based models like BERT are suited for classification and sentiment analysis due to their contextual understanding. Decoder-based models such as GPT-4 or LLAMA excel in text generation. Encoder-decoder models like BART are ideal for tasks requiring both comprehension and transformation, such as summarization or translation.

Apart from traditional prompt-based LLMs, LLM-based text embeddings can also be used in automated text analysis (Yoganarasimhan and Zheng 2025). Embeddings are dense, high-dimensional vector representations of text elements (like words, phrases, or sentences) that capture semantic information and enable prompt-based LLMs to understand relationships and meanings in language. For example, BERT can generate dynamic word embeddings such that the representation of a word changes based on the surrounding words. While embeddings are the initial step in transforming raw text into a numerical format suitable for processing by prompt-based LLMs, standalone LLM-based text embeddings can be used for tasks like semantic similarity, text-based segmentation, and topic modeling. Additional factors to consider when selecting the model include model size, cost, inference speed, and fine-tuning requirements.

Step 3: Prompt engineering. Prompt design critically influences output quality. Unlike traditional models, LLMs rely on well-structured prompts for task execution (Arora, Chakraborty and Nishimura, 2025). Prompts should be clear, specific, and include necessary context. For example, rather than using a general instruction like “Analyze this review,” a more effective prompt might be: “Analyze the sentiment of the following customer review for an online coffee shop. Is it positive, negative, or neutral? Highlight the key reasons.” Iterative refinement improves precision. For aspect-based sentiment analysis, prompting the model to address individual facets (e.g., “Rate the food, service, and ambiance”) yields targeted insights. Small changes in phrasing, using different markers (i.e., a, b, c vs. 1.2.3 vs. I, II, III), symbols, ordering of options, breaking tasks into sub-tasks, or specifying response format can significantly enhance accuracy (Reisenbichler et al., 2022).

Step 4: Select inference mode (zero, one/few-shot). The choice of inference mode affects model performance. *Zero-shot* requires no examples and works well for general tasks like sentiment classification. In-context learning models such as *one-shot* includes a single example to guide structure and format, while *few-shot* provides multiple examples, helping the model learn task-specific nuances. For instance, classifying mixed sentiment in reviews benefits from few-shot prompts that show how to parse praise and criticism in a single text. Balancing inference mode with task complexity and data availability ensures high-quality outputs. Next to the inference mode, additional layers of prompt framing can be used to increase reliability, consistency and comparability amongst inference modes. An example of such layers, amongst others, is RTF (Role – Task – Format) where the prompt is formatted in describing who the model is, what the task is and how the answer should be formatted (Li et al., 2024).

Step 5: Model evaluation and optimization (optional). While LLMs perform well out-of-the-box with well-designed prompts, systematic evaluation ensures output validity and consistency. Standard metrics such as accuracy or F1-score (see Table 3), along with human assessments, help verify coherence and factual alignment. For domain-specific tasks, fine-tuning on specialized data can improve alignment with terminology and context. Standard metrics together with loss and accuracy curves can be used to determine a balance between model optimization and prevention of overfitting. Alternatively, prompt tuning or lightweight adaptation techniques may enhance performance without the resource demands of full fine-tuning.

7. Enhancing text analysis in business research

This study addresses two persistent gaps in business research: the lack of conceptual frameworks to interpret business-relevant

communication and the underuse of modern tools for analyzing text data. Prior reviews have broadly focused on social media or marketing (e.g., Berger et al., 2020; Humphreys and Wang, 2017; Feuerriegel et al., 2025), but few have integrated communication theory with methodological guidance tailored specifically to business contexts. We contribute by applying message design logic (O’Keefe, 1988; O’Keefe and Lambert, 1995; Aakhus, 2007) to position business communication as a purposeful design activity in which message content, style, and structure reflect the communicator’s background, intentions, and context. We then connect this perspective to the approaches in text analysis to evaluate how different methods are suitable for different kinds of research inquiries (see Fig. 3 for an overview).

Our analysis of articles in the *Journal of Business Research* demonstrates how topic models, dictionaries, supervised machine learning, and large language models offer distinct but complementary strengths. Topic models are especially useful for exploratory analysis when categories are not predefined, allowing researchers to uncover latent themes (Lee, Yi, Kim, 2023; Lu et al., 2023). LDA offers a foundational approach, while models like BERTopic improve performance by capturing nuanced topic hierarchies (Jiang et al., 2023).

Dictionaries provide transparent, theory-driven tools for identifying predefined linguistic patterns (Guo et al., 2023; Wang et al., 2024a; Syrdal et al., 2023). They are particularly valuable when interpretability and replicability are essential. When dictionaries are not sufficient, researchers can refine or develop new ones, as demonstrated by studies applying domain-specific sentiment or behavioral constructs (Pan et al., 2023; Sun et al., 2024).

Supervised machine learning methods, such as logistic regression or support vector machines, are ideal for structured prediction tasks when labeled data are available for training (Filieri et al., 2023; Hajek et al., 2023). They balance predictive accuracy with interpretability, particularly when communicative features are well understood and constructs are easy to understand and code for (e.g., sentiment).

LLMs, such as BERT and GPT, significantly extend analytical capabilities (Dhakate and Joshi, 2023; Wang et al., 2024b). Encoder-based models like BERT are useful for classification and categorization; decoder-based models like GPT are suited for dynamic text generation; encoder-decoder models like BART are effective for summarization and rewriting.

Our review highlights that method selection should align with the research question, the structure of the data, and practical concerns such as transparency, data privacy, and resource constraints. For example, topic models are optimal for discovering and controlling for categories of content across documents in a data set, dictionaries support hypothesis-driven testing without labeled data, and SML or LLMs offer robust classification when objective labels are available.

Importantly, these methods are not mutually exclusive. Combinations often yield more powerful insights. For instance, topic models can identify recurring themes that inform the creation of dictionaries or training data for supervised models. LLMs can rapidly generate or label data, supporting initial exploration before deploying more interpretable tools (Li et al., 2024; Ye, Yoganarasimhan and Zheng, 2025).

We provide a checklist in Table 4 for to evaluate and advance text analysis quality in business research. This includes considerations for construct validity, method fit, interpretability, and evaluation rigor across topic models, dictionaries, SML, and LLMs. We also propose a future research agenda that maps key research objectives in research targeted at the *Journal of Business Research* to the four foundational premises of message design logic, namely the communicators’ background, context, intention, and impact, highlighting how each can be analyzed through text data using appropriate methods.

By integrating theory and methodology, this study supports researchers in extracting meaningful insights from text, providing both a conceptual rationale and a practical guide for aligning research questions with analytic tools. The broader implications of this work extend to disciplines such as organizational behavior, information systems, and

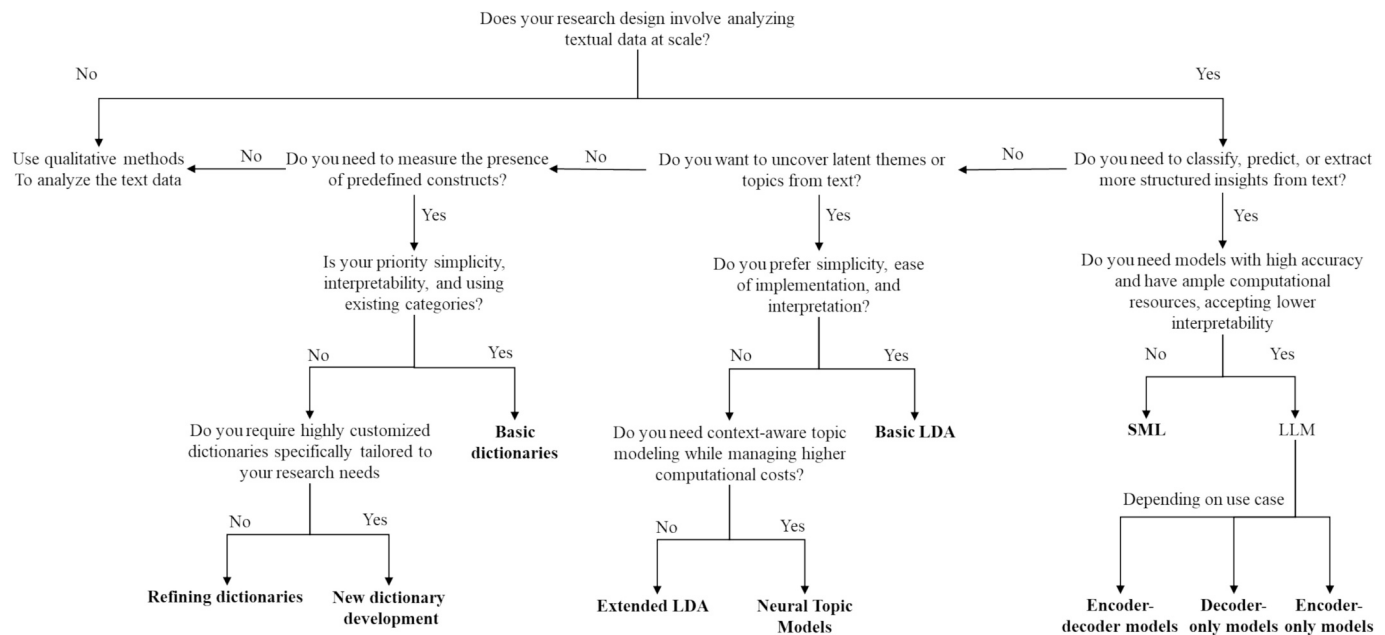


Fig. 3. Decision guide for choosing the right text analysis method. Note: LDA = Latent Dirichlet Allocation, SML = Supervised Machine Learning, LLM = Large Language Model.

Table 4
Checklist for authors and reviewers.

For all papers with textual data
✓ Is the conceptual approach towards the text data clearly stated (such as obtaining insights about the background of communicators, situation in which the communication occurs, intention behind the communication, or the impact of the communication)? ✓ Is the text data collection described transparently and sufficiently detailed? ✓ Is the (pre) processing of the raw text data described transparently and sufficiently detailed? ✓ Is the choice of the text-analytical method explained and justified? ✓ Is the choice of the text-analytical method appropriate, given the conceptual approach? ✓ Is the text-analytical method described transparently and sufficiently detailed? ✓ Are the assumptions and limitations of the chosen text-analytical method discussed?
For papers with topic models
✓ Are solutions for different the numbers of topics described and benchmarked? ✓ Is the choice of the solution used explained and justified, both empirically and conceptually? ✓ Does the chosen solution perform well enough regarding criteria such as coherence, stability, perplexity, and human evaluation? ✓ If needed: Is the chosen solution improved and fine-tuned? ✓ Is the chosen solution sufficiently described and, if possible, visualized?
For papers with dictionaries
✓ Is the focal construct of interest clearly defined and embedded in the relevant literature? ✓ Has the chosen dictionary been rigorously developed, either by the authors themselves or by the source of the dictionary? ✓ Is the dictionary appropriate for the specific research context and construct investigated? ✓ Is the measurement of the dictionary sufficiently converging with human perceptions of the text (e.g., accuracy, precision, recall, F1 score)?
For papers with supervised machine learning
✓ Is the feature engineering described transparently and sufficiently detailed? ✓ Is the labeling process described transparently and sufficiently detailed (i.e., labeling criteria, training of human coders, labeling process, inter-coder reliability, iterative refinement)? ✓ Is the choice of the machine learning model explained and justified? ✓ Does the chosen machine learning model perform well enough regarding criteria such as accuracy, precision, recall, F1 score, and AUC-ROC? ✓ If needed: Is the chosen machine learning model solution improved and fine-tuned?
For papers with large language models
✓ Is the construct of interest clearly defined and aligned with the prompt design or fine-tuning objective? ✓ Is the prompt engineering process described transparently and sufficiently detailed (e.g., zero-, one-, or few-shot setup, prompt iterations, task framing)? ✓ Is the choice of the LLM architecture (e.g., encoder-only, decoder-only, encoder-decoder) explained and justified based on the research question and task? ✓ Does the LLM output perform well according to relevant evaluation criteria such as accuracy, F1-score, or human evaluation for interpretability and relevance? ✓ If needed: Is the model fine-tuned, adapted (e.g., via prompt tuning), or iteratively refined, and is this process described transparently?

communication studies, where understanding stakeholder intentions, perceptions, and interactions through text is equally essential.

8. Future business research directions with text data

8.1. Using text analysis to understand the background of communicators

Future research should deepen our understanding of how communicator backgrounds are reflected in their business communication. Text analysis offers the ability to uncover linguistic indicators of stakeholder

characteristics, whether from consumers, employees, or organizations. For example, customer-generated texts may reflect different stages of the customer journey, providing insights relevant to consumer behavior. Text can also reveal unspoken expectations or needs, which may inform service design, product development, or policy interventions, particularly in service and innovation research. As digitization expands access to historical data, researchers can use corpora like the New York Times Annotated Corpus or [Amazon.com](https://www.amazon.com) to trace how attitudes and preferences have evolved over time. Especially change-based models might become valuable to understand quasi-causal relations between text and business-relevant outcomes over time.

8.2. Using text analysis to understand the situation of communicators

Context remains central to how communicators design their messages. Future studies should investigate how relational, business or societal relevant situational factors influence communicative choices. For example, analyzing consumer-generated content can reveal how public attitudes become polarized. Non-verbal and environmental factors from settings like customer-frontline employee relations in service chats and or organizational circumstances can offer valuable settings for studying how context is reflected in message design and in turn message design can then serve as a proxy for such circumstances. Future work could also explore how different communication contexts enable or constrain the expression of relational cues, including interactions with non-human agents such as avatars or voice assistants.

8.3. Using text analysis to understand the communicators' intentions behind communication

Future research should also examine how text reflects the communicators' intentions. Business texts often contain cues to communicative goals, for example, whether to inform, persuade, or prompt action. For instance, analyzing firm-generated texts may reveal how communication goals evolve across the customer journey, providing insights relevant to targeted advertising, marketing communications, and customer engagement. Researchers should also study how communication between firms and stakeholders signals their strategic intentions, such as initiating collaboration, investment, or market expansion. Investigating how text analysis can support the detection of malicious content, while safeguarding freedom of expression, presents an important ethical dimension with implications for corporate social responsibility and business ethics.

8.4. Using text analysis to understand the impact of communication on receivers

Finally, understanding how communication affects recipients remains a critical avenue for future business research. Text analysis enables the development of models to predict how content, style and structure of messages influence audience perception, judgment, and action. Researchers should continue to investigate how message framing or content themes influence decisions. For example, at various stages of the consumer journey, contributing to work on consumer behavior and wellbeing. Retail and multichannel management research would benefit from studying how content formats vary in effectiveness across platforms. Text analysis can also inform communicative strategies to promote more mindful and sustainable consumption, an increasingly important area in business ethics. In business-to-business contexts, future work should identify communicative signals that reduce risk, improve efficiency, or foster trust. These insights would valuable add to the current understanding of inter-organizational collaboration, exchange performance, and the link between communication and strategic outcomes in business-to-business marketing and strategic management.

8.5. Future research on text analysis methods

Future research on text analysis methods should address several emerging priorities that build directly on the advancements of these methods. One important direction is the comparison between domain-specific models and general-purpose models (e.g., GPT-4). Future work should examine when specialized models deliver meaningful improvements over general models. Such studies would help clarify trade-offs between performance, cost, and accessibility. Another key area involves the tension between model complexity and interpretability, particularly with topic models and LLMs. Comparing the utility of transparent approaches (e.g., dictionaries) against more complex black-box systems can guide more deliberate methodological choices and inform practical implementation. If more advanced LLMs are used then research should help establish guidelines as to how results can be reliably reproduced even though the technology advances.

The reliability and accuracy of different text mining approaches across multilingual, and multimodal contexts is another interesting research direction. As business research becomes increasingly global, future studies should investigate the reliability of multimodal and multilingual text analysis approaches. Particular attention should be given to cultural variation in how messages are reflective of communicators with different backgrounds and how multimodal and multilingual features impact receivers. Future studies should also investigate the implications of bias, transparency, and data privacy in automated text analysis, especially when using LLMs. Clear ethical frameworks are needed to guide responsible use, particularly in sensitive domains or high-stake decisions. Finally, researchers need to be mindful that not every dataset can be uploaded and mined with a publicly available LLM if the data is sensitive.

CRedit authorship contribution statement

Dennis Herhausen: Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Stephan Ludwig:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Ehsan Abedin:** Writing – review & editing, Writing – original draft, Software, Methodology. **Nasim Ul Haque:** Writing – review & editing, Writing – original draft. **David de Jong:** Writing – review & editing, Writing – original draft.

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